DEPARTMENT OF ECONOMICS
AND BUSINESS ECONOMICS
AARHUS UNIVERSITY

EXAM ASSIGNMENT

Study Programme and level	**MSc Operations and Supply Chain Analytics**						
Term	Winter 2025/26 - ordinary exam						
Course name and exam code(s)	Tools for Analytics						460202E016
Exam form and duration	Written onsite exam, internet allowed						3 hours
Date and time	13 January 2026						10:00-13:00
Supplementary material/aids	All	X	Specified		No		
Anonymous exam	Yes	X	No		Please do **not** write your name or student ID number anywhere.		
Use of generative AI (GAI) allowed	Yes		No	X			
Hand-in of handwritten material allowed	Yes		No	X			
Hand-in of extra material (appendix) in WISEflow allowed	Yes	X	No				
Other relevant information	**Avoid being suspected of exam cheating** Remember to state references and use quotation marks, if you copy text from other sources or re-use parts of a previously submitted exam paper (plagiarism and self-plagiarism). Students must answer the exam assignment **individually**. All submitted exam papers are checked for plagiarism, so cheating and collaboration between students will be detected.						
Number of pages (incl. front page)	6 pages						

Other instructions:
It is important that you start uploading your exam paper well in advance – at least 10 min. before end of exam.

Practical information

- This exam is with internet. You may use any online or book-based resource you would like, but you must include citations for any code that you use (directly or indirectly). You may not consult with anyone else during this exam. That is, you cannot ask direct questions on the internet, use answers given on the internet after the start of this exam, or consult with each other, not even for hypothetical questions.
- The use of tools based on Generative AI, such as ChatGPT or Copilot, is not allowed.
- This assignment has an appendix available for download from WISEflow.
- Please note that the weights on each assignment are only guideline weights, and that they only provide information regarding the relative weight of the assignments. The final evaluation will be given based on the total material handed in.
- If you find that some information is missing in the assignments, you may make the necessary assumptions and clearly specify these.
- Handing in: You must hand in a set of files (`.Rmd` and `.html`) as "Appendix material". Due to the system, you must also upload an empty PDF document named `<participant index>.pdf`.
- Your R code will be tested using R version 4.5.2 (2025-10-31).
- Your R code must be written in an R Markdown (Rmd) file named `<participant index>.Rmd`. Moreover, hand in the rendered/knitted HTML file `<participant index>.html`.
- Your file must include your code and a (brief) comment for each question. For example, "The three companies with the smallest profit are ..." or "The plot shows that ...".
- An R Markdown template file is given in the appendix on WISEflow, which you may use as a starting point.
- You may load and use the following packages:

  ```` r
  library(tidyverse)
  library(skimr)
  library(conflicted)
  ```
  ```
  ```

## R - Assignment 1 (55%)

Consider the dataset in the file `catch.csv` containing catches from rivers in Denmark. A row is given for each catch with information about weight, length, date, angler, place, method etc.

Some of the columns/variables are:

- `Date`: Date of catch.
- `Length`: Length of fish (cm).
- `Weight`: Weight of fish (kilograms).
- `Place`: Place of catch (Lower, Middle, Upper).
- `Method`: Method used (Spin, Fly, Worm).
- `Killed`: If the fish was killed.
- `River`: River of catch.
- `Species`: Species of fish.
- `Name`: Name of angler.

The dataset can be read using:

```
library(tidyverse)
dat <- read_csv("catch.csv")
```

Moreover, consider the dataset in the file `weight.csv` containing weight estimates for a given length. A row is given for each length with information about weight, month, river, species etc.

Some of the columns/variables are:

- `Length`: Length of fish (cm).
- `Month`: Month considered.
- `Avg`: Average weight of fish (kilograms).
- `Lower`: Lower confidence interval weight (kilograms).
- `Upper`: Uppper confidence interval weight (kilograms).

The dataset can be read using:

```
library(tidyverse)
dat_weight <- read_csv("weight.csv")
res <- dat_weight |>
   filter(River == "Karup",
          Species == "Seatrout",
          Length == 50,
          Month == 5) |>
   print()
```

```
## # A tibble: 1 x 7
##   Length   Avg Lower Upper Month River Species
##    <dbl> <dbl> <dbl> <dbl> <dbl> <chr> <chr>
## 1     50  1.54  1.24  1.92     5 Karup Seatrout
```

That is, for instance, the average weight of a 50 cm seatrout in Karup River in May is 1.54 kg with a 95% confidence interval of [1.24, 1.92] kg.

Use the *dplyr* package in *tidyverse* to calculate relevant summary tables (tibbles/data frames), and answer/complete the following questions/tasks:

### Question 1

Which method resulted in most fish caught in Skjern River?

### Question 2

Which angler has caught most fish in total and in kilograms over all the years considered?

### Question 3

Calculate the condition factor for each fish using the formula `Weight*100000/Length^3`.

Is the average condition factor higher for salmon or seatrout?

Consider the months after April and fish with a weight above 50 cm. Do seatrout in Karup River have a higher average condition factor than seatrout in Skjern River?

### Question 4

Are there more catches per day on average on weekdays or during weekends in Skjern River?

### Question 5

Calculate the catch and release rate (number released / total number caught) for each river over all years. Have the catch and release rates increased over the years in Karup River?

### Question 6

Use the weight dataset to update missing weights in the catch dataset. That is, for each fish, with a missing weight, use the average weight from the weight dataset for the corresponding length, river, species and month. Hint: Before joining the datasets, you need to round the length to an integer.

## R - Assignment 2 (45%)

Answer this assignment using the *ggplot2* package in *tidyverse* (you may need *dplyr* for preparing the datasets you want to plot). We use the same dataset as given in Assignment 1 if not specified.

### Question 1

Create a visualization showing the average weight of seatrout in Karup River over the months with the following features:

- Use the dataset from Assignment 1 in which missing weights have been updated if possible.
- Bars with the fill colour `darkred` are used.
- An informative figure title and axis titles are given.

Which month has the highest average weight?

### Question 2

Create a visualization showing the relationship between length and weight for all fish caught with the following features:

- Different colours are used for each species.
- Points are used to show the individual catches with some transparency (alpha).
- A trend line is added for each species.
- An informative figure title and axis titles are given.
- The legend is placed at the bottom of the figure.

Are salmon generally heavier than seatrout for a given length? Can salmon reach a higher weight than seatrout?

### Question 3

Consider the weight dataset. Create a visualization showing the weight for a given length with the following features:

- Only Skjern River is considered.
- Lines are used to show the average weight given length for each species.
- Different colours are used for each month.
- A subplot is made for each species.
- An informative figure title and axis titles are given.
- The legend is placed at the bottom of the figure.

In which month is the weight lowest for seatrout?

### Question 4

Create a visualization showing the percentage of fish over 59 cm killed over the years with the following features:

- Bars are used to visualize the percentage of fish killed.

- Different fill colours are used for each river (use `position = "dodge2"` to place bars beside each other).
- An informative figure title and axis titles are given.
- The legend is placed at the bottom of the figure.

In which river do most anglers catch and release fish (i.e. the lowest percentage killed)?

### Question 5

Consider seatrout. For each river, year, and month, calculate the number of catches. Next, calculate the catch percentage for each month (number caught per month / total number caught per year).

Create a visualization showing the variation in these percentages for each month with the following features:

- A boxplot is used to show the variation of the catch percentages.
- A subplot is made for each River.
- An informative figure title and axis titles are given.

Comment on the results for July.